Foreground Refinement Network for Rotated Object Detection in Remote Sensing Images

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***Abstract*— Object detection has been a fundamental task in the field of remote sensing and has made considerable progress in recent years. However, the high background complexity in remote sensing images (RSIs) remains challenging. In this article, we propose a refined rotation detector, namely, the Foreground Refinement Network (FoRDet), to alleviate the above problem by leveraging the information of foreground regions from the perspectives of feature and optimization. Specifically, we propose a foreground relation module (FRL) that aggregates the foreground-contextual representations from the coarse stage and improves the discrimination of foreground regions on feature maps in the refined stage. Besides, considering the risk of the potential foreground anchors being overwhelmed in the training phase, we design a foreground anchor reweighting (FRW) loss that integrates the classification confidence and localization accuracy of each foreground anchor from the coarse stage to dynamically regulate their contributions in the refined stage, which highlights the potential foreground anchors. The comprehensive experimental results on three public datasets for rotated object detection DOTA, HRSC2016, and UCAS-AOD demonstrate the effectiveness of our proposed method.**

***Index Terms*— Deep learning, foreground modeling, remote sensing images (RSIs), rotated object detection.**

I. Introduction

With the rapid development of the high spatial resolution earth observation technique, the automated interpretation of remote sensing images (RSIs), including scene classification, object detection, and road extraction, has been an active topic [1]–[8]. Object detection, as a fundamental task in the field of RSIs, aims to accurately localize and identify the object of interest (e.g., airplanes, vehicles, and ships) and has been widely used in traffic monitoring, search and rescue missions, and urban planning. Compared with the natural scene, RSIs are typically taken with bird views and acquired from different areas of the earth with variant ground sampling distances (GSDs) making object detection is more challenging. The main reasons are given as follows.

1) *Arbitrary Orientations:* The bird views of RSIs lead to the various orientations of geospatial objects, making it difficult to regress the orientations.
2) *Huge Scale Variations:* The huge scale variations among objects from different categories require the detector to simultaneously handle extremely tiny and huge objects.
3) *High Background Complexity:* The foreground regions are always accompanied by the complex background, which may seriously influence the feature extraction of foreground regions.



(a) (b)

Fig. 1. (a) High background complexity in RSIs. (b) Risk of potential foreground anchor (i.e., the blue one) being overwhelmed.

Benefiting from the recent advances of object detection in the natural scene [9]–[13], many researchers extended the off-the-shelf detectors into the RSIs and proposed various effective approaches to handle the scale variations and arbitrary orientations of geospatial objects [14]–[19]. However, the high background complexity is still a tough problem for object detection in RSIs. As sketched in Fig. 1(a), the RSIs may have complex backgrounds. For example, some vehicles are blocked by the shadows of trees or buildings, and the background noises may have a similar rectangular appearance to vehicles. These complex backgrounds may bring noises for foreground regions, limiting the detection performance. In addition, since the existing detectors treat all foreground anchors (also known as positive anchors) equally, the potential foreground anchors may be overwhelmed in the training phase.

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As shown in Fig. 1(b), although the IoU values between all the foreground anchor points and the ground truth (i.e., the red box) are greater than the positive threshold, the blue one is more likely to contain the foreground region. Thus, we should distinguish different foreground anchors to put more focus on the potential foreground anchors.

In this article, we propose a refined rotation detector, called the Foreground Refinement Network (FoRDet), to alleviate the interference of complex background in RSIs. Because of the high-precision detection performance of the refined object detectors [19]–[21], our FoRDet follows the refined rotation structure and proposes the foreground relation module (FRL) and foreground anchor reweighting (FRW) loss to enhance the foreground regions. For the FRL, we present a relation mechanism [22]–[24] to aggregate the foreground-contextual information in the coarse stage and enhance the foreground regions on the feature maps in the refined stage. For the FRW loss, we integrate the classification confidence and localization accuracy from the coarse stage to highlight the potential foreground anchors in the refined stage. Besides, since the proposed loss function increases the loss weights of foreground anchors, it also eases the foreground-background imbalance to a certain extent.

Our main contributions can be summarized as follows.

1) We propose the FoRDet that leverages the information of foreground regions from the perspectives of features and optimization in order to alleviate the problem of high background complexity in RSIs.
2) To enhance the discrimination of the foreground region on the feature maps, we propose the FRL module that aggregates the foreground-contextual representations under the supervision of the foreground regions.
3) To highlight the potential foreground anchors in the training phase, an FRW loss is designed to dynamically regulate the loss weights of foreground anchors.

The remainder of this article is organized as follows. Section II gives a brief introduction to the related work. In Section III, we describe our proposed method in detail. We report and discuss the experiments in Section IV. Finally, Section V concludes this article.

II. Related Works

A. Object Detection in Remote Sensing Community

Object detection in RSIs has been extensively studied for many years. Most earlier works [25], [26] employ handcrafted features, yielding poor performance. Benefiting from the advancement of CNN, many researchers proposed outstanding detectors and facilitated the field of geospatial object detection to flourish. RICNN [27] introduces a new rotation-invariant layer on the existing CNN architectures to tackle the problem of geospatial object rotation variations. Deng *et al.* [15] adopted the concatenated ReLU and Inception module into the feature extractor to increases the variety of receptive field sizes and designs a multiscale object proposal network for a different region scale generation. Different from the existing multiscale object detection approaches, Zheng *et al.* [18] devised the HyNet to learn the hyperscale feature representation, thus alleviating the extreme scale-variation problem in RSIs. However, since the geospatial objects have arbitrary orientations, the abovementioned horizontal bounding box (HBB)-based methods lead to the mismatch between the HBB and rotated objects [16]. Thus, many studies introduced the oriented bounding boxes (OBBs) to eliminate the mismatch problem. Considering the misalignment between horizontal RoI features and objects, Ding *et al.* [16] designed the RoI-Transformer, a lightweight supervised RoI-wise operation, to efficiently model the geometry transformation and relief the misalignment. Yang *et al.* [19] proposed the R^3Det that predicts the OBB from the horizontal default anchor through two-stage regressions. Xu *et al.* [17] characterized multioriented objects by introducing the relative gliding offset of four vertexes of the HBB, which demonstrates superior performances on multiple datasets. Moreover, some anchor-free methods [28]–[31] have been presented, which avoids devising complex rotated anchors and achieves comparable performance. Our method concentrates on the challenge of the high background complexity in RSIs and alleviates this problem by aggregating the information of foreground regions. Considering the good tradeoff in speed and accuracy of R^3Det [19], our FoRDet also leverages a refined rotation structure to predict the objects at arbitrary orientations.

B. Foreground Modeling

The complex background is one of the challenges in RSIs, and several studies have shown that modeling the foreground information can improve the performance in scene classification [32], [33], geospatial object detection [34]–[37], and geospatial object segmentation [38], [39]. Zhu *et al.* [32] used the Grad-CAM [40] to generate the attention map of the input image and fed it into the network as an additional input. With the help of the attention map, the network focuses more on foreground discriminative regions. Seeing the lack of foreground modeling in general semantic segmentation methods, Zheng *et al.* [38] proposed a foreground-aware relation network for geospatial object segmentation. The proposed network aims to learn the foreground-scene relation and uses the learned relation to enhance the foreground features. Feng *et al.* [37] designed a global context-aware enhancement module to adaptively strengthen the features of high-quality foreground proposals and suppresses the features of cluttered backgrounds, which achieves remarkable performance in weakly supervised object detection for RSIs. Yang *et al.* [34] adopted a supervised learning way to enhance the foreground object information in geospatial object detection and proposed the pixel attention module. The pixel attention module predicts the foreground saliency map under the guidance of foreground supervision and fuses the prediction results with network features to strengthen the activation of foreground regions. Similar to [34], RECNN [35] introduces a supervised saliency reconstruction network and multilayer fusion strategy to strengthen the feature representation of object regions. Contrast to previous works [32], [33], [36]–[39], we capture the foreground-aware relation in RSIs with a supervised learning approach. In addition, compared to the methods [34], [35] that generate foreground saliency maps as supervision, we design a more concise way to introduce foreground information guidance.

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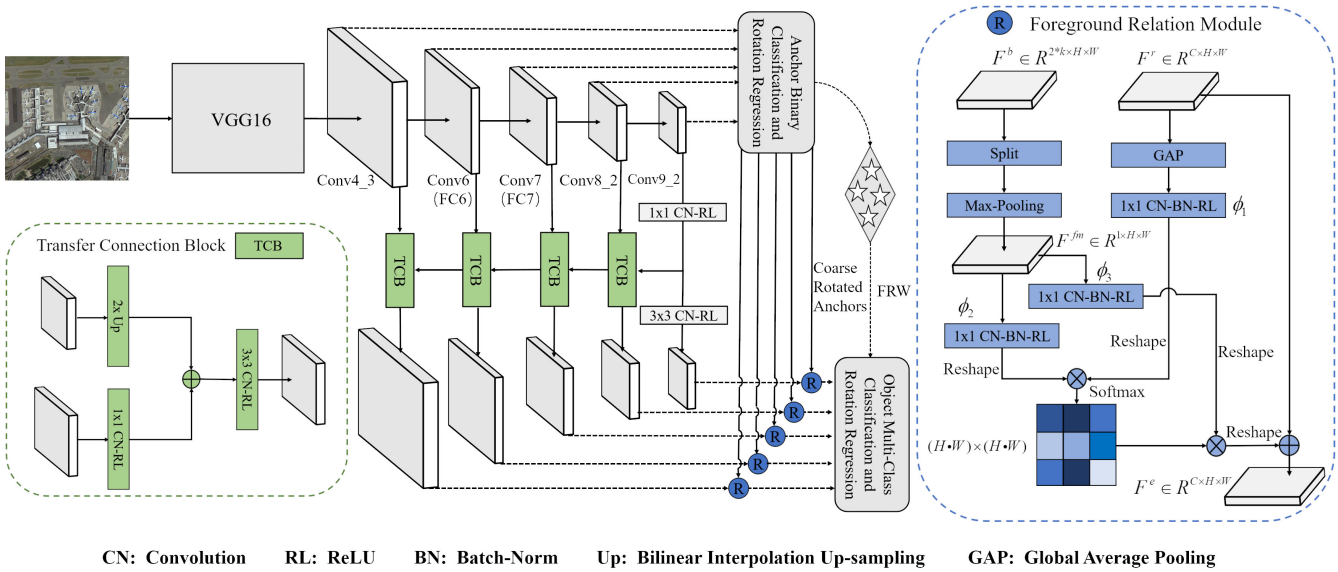


Fig. 2. Overall framework of our FoRDet. The refined rotation structure provides a good tradeoff in speed and accuracy for rotated object detection. The FRL leverages the foreground-contextual information to enhance the discrimination of the foreground regions. The FRW loss dynamically adjusts the loss weight of different foreground anchors in the training phase.

C. Reweighting Mechanism

Due to the differences between training samples, many researchers [41]–[46] have designed the reweighting mechanism to distinguish the importance of different samples and achieved promising results. Considering the severe imbalance between large and small objects in the same RSI, FMSSD [41] proposes an area-weighted loss to make the network focus on small objects. For each positive RoI, its loss weight is related to the area of its corresponding ground truth. If the ground-truth area is small, a larger weight is applied to the RoI. Yan *et al.* [42] designed the IoU-based weighted loss in both the classification and detection branch to assign the loss weight of each positive RoI according to their IoU values. Sun *et al.* [43] proposed an adaptive saliency biased loss (ASBL) to improve the detection performance in RSIs. The ASBL estimates the saliency information of input images and anchors with the pretrained CNNs and assigns different loss weights to each training image and anchor based on its saliency estimation. Ming *et al.* [44] combined the IoU before and after the regression of each anchor to dynamically assign each anchor and adjust their loss weight, which can select high-quality anchors. Different from the above works, we focus on the difference between the foreground anchors and devise an FRW loss that comprehensively considers the classification confidence and the localization accuracy of each foreground anchor to dynamically adjust its loss weights.

III. Proposed Method

Our FoRDet is based on a refined rotation structure and introduces the FRL and FRW loss to strengthen the foreground regions. The overall framework of our network is illustrated in Fig. 2.

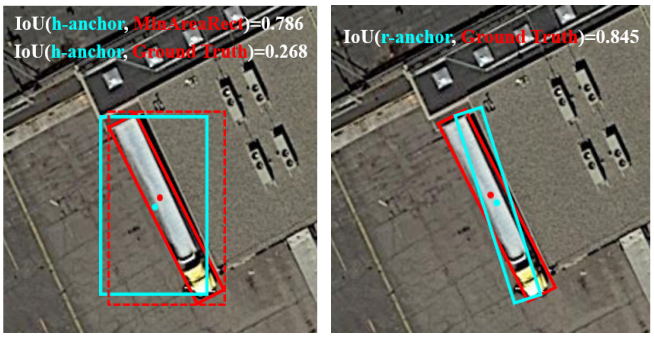


Fig. 3. Different default anchor settings and matching strategies in (a) coarse stage and (b) refined stage. (a) IoU between the horizontal anchor and the minimum bounding rectangle of the ground truth. (b) Skew IoU between the coarse rotated anchor and the ground truth.

A. Refined Rotation Detector

Benefiting from the superiority of the two-stage detector, some researchers add one or more refined stages to the one-stage detectors to obtain better performance [19]–[21]. Inspired by these previous works, our FoRDet also has a refined rotation structure for high-precision rotated object detection. Different from RefineDet [20], the proposed method employs the horizontal anchors and the rotated anchors in the coarse stage and the refined stage, respectively, to achieve better performance for the rotated object detection. Compared to [19] and [21], we devise the FRW loss from the perspective of optimization to highlight the potential foreground anchors.

Our FoRDet can be divided into two stages. In the coarse stage, we leverage the horizontal anchors, which can be regarded as the rotated anchors with an angle of $-\pi/2$, as the default anchors. For the matching strategy, we calculate the IoU between the horizontal anchor and the minimum bounding

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rectangle of the ground truth [see Fig. 3(a)]. The advantages of the above setting lie in the horizontal anchors provide a higher recall compared to the rotated ones and can avoid the computational complexity caused by the design of rotated anchors with different angles [47]–[49]. In the refined stage, we set the coarse rotated anchors, obtained in the coarse stage, as the default anchors to predict more precision rotated bounding boxes. The rotated anchors avoid the interference of background noise contained in the horizontal anchors. Thus, the refined stage can ensure higher precision. Different from the coarse stage, we use the skew IoU [47] between the coarse rotated anchor and the ground truth to select the foreground anchors [see Fig. 3(b)].

For the definition of rotated objects, we use the five-parameter (x, y, w, h, θ) method [16], where $\theta \in [-\pi/2, \pi/2)$ denotes the angle from the direction of the x-axis to the direction of the width w. Besides, we add an angular offset to the regression branch of both two stages to predict the rotated bounding box.

B. Transfer Connection Block

The transfer connection block (TCB) has two main functions. On the one hand, the TCB converts features of the different layers from the coarse stage into the refined stage, establishing the connection between the two stages. On the other hand, the TCB propagates the rich semantic information from the high-level features to the low-level ones through a top-down path, which can improve the detection performance.

Fig. 2 shows the detailed structure of the TCB, and it has two input feature maps with different spatial resolutions. For the low-resolution feature map, the nearest bilinear up-sampling operation is adopted to up-sample the spatial resolution by a factor of 2. The high-resolution one undergoes a 1×1 convolution layer to reduce the channel dimension to 256 and is then merged with the up-sampled low-resolution feature maps by elementwise addition. Finally, the merged feature map is fed into a 3×3 convolution layer to generate the final feature map.

C. Foreground Relation Module

Due to the high background complexity in RSIs, it is difficult for the model to distinguish the foreground regions from the background. To solve this problem, we devise a conceptually simple yet effective FRL to improve the discrimination of the foreground regions on the feature maps by exploiting the foreground-contextual representations. Different from [22] and [24], the FRL module employs a supervision way to better learn the foreground-contextual representations.

1) Foreground Map Generation: In our refined rotation structure, the output feature map $F^b \in R^{2*k\times H\times W}$ (k represents the number of anchors in each location on the feature map) of the binary classification branch in the coarse stage predicts each anchor as foreground/background under the supervision of the ground-truth category, which can be used to generate the foreground map that strengthens the potential foreground regions. Thus, for each location on the feature map F^b, we first split the foreground features of all anchor points at this location and then employ the max-pooling operation along the channel axis on the split foreground features to generate the foreground map: $F^{\text{fm}} \in R^{1\times H\times W}$.

2) Foreground Relation Representations: Similar to [22]–[24], we employ an embedding dot production to represent the foreground relation, as defined in the following equation:

$$\text{FR}(F^r, F^{\text{fm}}) = f(\phi_1(F^r)^T \cdot \phi_2(F^{\text{fm}})) \tag{1}$$

where $F^r \in R^{C\times H\times W}$ is the feature map in the refined stage and F^{fm} is the generated foreground map. ϕ_1 and ϕ_2 denote two embedding functions.

For the feature map F^r, we first use the global average pooling (GAP) to reduce the channel dimension and then feed it into a convolution layer with 1×1 filters to learn the embedding feature map. Thus, ϕ_1 can be formulated as

$$\phi_1(F^r) = \text{Conv}_{1\times 1}(\text{GAP}((F^r)). \tag{2}$$

For the foreground map F^{fm}, we employ a 1×1 convolution operation as the embedding function

$$\phi_2(F^{\text{fm}}) = \text{Conv}_{1\times 1}((F^{\text{fm}})). \tag{3}$$

After obtaining the two embedding feature maps, we reshape $\phi_1(F^r)$ and $\phi_2(F^{\text{fm}})$ into $\text{HW} \times 1$ and $1 \times HW$, respectively, and adopt the softmax operation to calculate the foreground relation $\text{FR}(F^r, F^{\text{fm}}) \in R^{HW\ \times \text{HW}}$

$$\omega_{ij} = \frac{\exp\left(\phi_1\left(F_i^r\right)^T \cdot \phi_2\left(F_j^{\text{fm}}\right)\right)}{\sum_{j=1}^{\text{HW}} \exp\left(\phi_1\left(F_i^r\right)^T \cdot \phi_2\left(F_j^{\text{fm}}\right)\right)} \tag{4}$$

where ω_{ij} measures the relation between the ith pixel of the feature map F^r and the jth pixel of the foreground map F^{fm}.

3) Foreground-Contextual Aggregations: To aggregate the foreground-contextual information to the feature map in the refined stage, we first feed F^{fm} into a 1×1 convolution layer to generate the embedding feature map and reshape it into $R^{\text{HW}\ \times 1}$. Next, we integrate the foreground relation representations and the embedding feature map by a matrix multiplication to obtain the foreground context and reshape it to $R^{H\times W\times 1}$. Finally, we employ the elementwise sum operation to aggregate the foreground context with F^r, and the final output F^e can be calculated as follows:

$$F_i^e = \sum_{j=1}^{\text{HW}} \omega_{ij} \cdot \phi_3\left(F_j^{\text{fm}}\right) + F_i^r \tag{5}$$

where ϕ_3 denotes the 1×1 convolution layer.

D. Foreground Anchor Reweighting Loss

In order to prevent the potential foreground anchors from being overwhelmed in the training phase, we propose an FRW loss in the refined stage to highlight them. At the same time, due to the emphasis on the foreground anchors in the FRW loss, the foreground-background imbalance problem can be alleviated by some degrees.

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1) Foreground Scores: First, we define the foreground score to distinguish the difference between each foreground anchor. For each foreground anchor in the refined stage, we leverage its classification and regression information from the coarse stage to define its foreground score.

For the classification perspective, we use the foreground probability of each foreground anchor predicted from the coarse stage as the classification confidence. From the regression aspect, the localization accuracy is based on the IoU of each foreground anchor and its corresponding ground truth. Thus, the foreground score of a foreground anchor a is written as follows:

$$\mathrm{FS} = \alpha \cdot \mathrm{loc}_a + (1-\alpha) \cdot \mathrm{cls}_a \tag{6}$$

where loc_a and cls_a are the localization accuracy and the classification confidence, respectively, and α is a balance parameter to adjust the impact of localization and classification.

2) Reweighting Factor: Since the foreground score reflects the possibility that each foreground anchor contains foreground objects, we propose to dynamically regulate the weight of each foreground anchor in the refined stage according to their foreground scores. Previous studies [41], [50] have shown that a large variance among the weights of different anchors leads to better performance. Thus, we introduce an exponential function to transform the foreground score to the reweighting factor r

$$r = \alpha \cdot \exp(\beta \cdot \mathrm{loc}_a) + (1-\alpha) \cdot \exp(\beta \cdot \mathrm{cls}_a) \tag{7}$$

where β is used to further enlarge the variance.

3) Dynamic Training Strategy: In the initial period of the training phase, the performance of the coarse stage is poor and may cause the reweighting factors unconfident in the refined stage. These unconfident reweighting factors may lead to instability of model training and even make the model unable to converge. To alleviate this problem, we devise a dynamic training strategy [38], [51] that leverages an annealing function $\Gamma(t)$ related to the training iteration to gradually switch ON the reweighting mechanism. Thus, the reweighting factor under the dynamic training strategy can be written as follows:

$$r = \alpha \cdot \exp(\Gamma(t) \cdot \beta \cdot \mathrm{loc}_a) + (1-\alpha) \cdot \exp(\Gamma(t) \cdot \beta \cdot \mathrm{cls}_a) \tag{8}$$

$$\Gamma(t) = \begin{cases} (t/T)^2, & \text{if } t < T \\ 1, & \text{if } t >= T \end{cases} \tag{9}$$

where t is the current training iterations and T is the annealing steps. In all subsequent experiments, we set T to 1/12 of the total training iterations.

In summary, the FRW Loss in the refined stage can be described as

$$\mathcal{L}_{\mathrm{FRW}} = \frac{1}{N_{\mathrm{refined}}} \left(\sum_i r_i \mathcal{L}_m\left(c_i, l_i^*\right) + \sum_i \left[l_i^* \geq 1\right] r_i \mathcal{L}_r\left(t_i, g_i^*\right) \right) \tag{10}$$

where N_{refined} indicates the number of anchors in the refined stage. i is the index of anchor in one batch in the refined stage. r_i is the reweighting factor of anchor i. c_i and l_i^* are the predicted probability and the ground-truth label of anchor i. $[l_i^* \geq 1]$ is the Iverson bracket indicator function. t_i denotes a vector representing (x, y, w, h, θ) of the predicted rotated bounding box and g_i^* is that of the ground truth. We adopt the cross-entropy loss over multiple classes and the smooth L1 loss [9] as the multiclass classification loss $\mathcal{L}_m$ and the regression loss $\mathcal{L}_r$, respectively.

E. Loss Function

The loss function for our FoRDet consists of two parts, i.e., the loss in the coarse stage and the FRW loss in the refined stage. The coarse stage assigns a binary class label (of being foreground or not) to each anchor and regresses its location to generate the coarse rotated anchor. For the refined stage, it further predicts object categories and the accurate rotated bounding box. Thus, the loss function can be defined as

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\mathrm{coarse}} + \mathcal{L}_{\mathrm{FRW}} \\ &= \frac{1}{N_{\mathrm{coarse}}} \left(\sum_i \mathcal{L}_b\left(p_i, \left[l_i^* \geq 1\right]\right) + \sum_i \left[l_i^* \geq 1\right] \mathcal{L}_r\left(x_i, g_i^*\right) \right) \\ &\quad + \frac{1}{N_{\mathrm{refined}}} \left(\sum_i r_i \mathcal{L}_m\left(c_i, l_i^*\right) + \sum_i \left[l_i^* \geq 1\right] r_i \mathcal{L}_r\left(t_i, g_i^*\right) \right) \end{aligned} \tag{11}$$

where N_{coarse} represents the numbers of anchors in the coarse stage. p_i and x_i are the predicted probability of the anchor i being an object and regressed rotated coordinates of the anchor i in the coarse stage. Similar to RefineDet [20], the cross-entropy loss over two classes (foreground versus background) is employed as the binary classification loss $\mathcal{L}_b$.

IV. Experiments

A. Datasets

1) DOTA [52]: It is a large-scale dataset for object detection in RSIs with OBB annotations, which contains 2806 images and 188 282 geographic objects. These geographic objects are mainly divided into 15 categories, including plane (PL), baseball diamond (BD), bridge (BR), ground field track (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). The whole dataset is randomly divided into three parts, of which 1/2 as the training set, 1/6 as the validation set, and 1/3 as the test set.

Since each image in the DOTA dataset is sized about 4000×4000 pixels, we crop the original image into 1024×1024 patches with an overlap of 200 pixels by the officially provided toolkit[1] and scale each patch to 512×512. With these processes, we extract 10 276 patches for training, 2957 patches for validation, and 10 833 patches for testing. The model is trained by 180k iteration in total where the learning rate starts from 1e-3 and decreases to 1e-4 and 1e-5 at the 120k iterations and 150k iterations.

[1] https://github.com/CAPTAIN-WHU/DOTA_Devkit

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TABLE I
ANCHOR SCALE SETTINGS IN EACH LAYER

Layer_name	Stride	Min_size	Max_size	Aspect_ratio
Conv4_3	8	20	51	0.5, 1, 2
Conv6	16	51	184	0.5, 1, 2
Conv7	32	184	317	0.5, 1, 2
Conv8_2	64	317	450	0.5, 1, 2
Conv9_2	128	450	583	0.5, 1, 2

2) HRSC2016 [53]: The HRSC2016 dataset is a challenging arbitrary orientation ship detection dataset. The dataset is collected from Google Earth and includes images from two scenarios: ships on the sea and ships close inshore. The image resolution range from 2 to 0.4 m and the image size are between 300×300 and 1500×900. The training and test sets include 617 images and 444 images, respectively. For the experiments on the HRSC2016 dataset, we also scale the image to 512×512 for training and testing. We train the model with 1e-3 for the first 100k iterations and then 1e-4 and 1e-5 for the other two 10k iterations.

3) UCAS-AOD [54]: The UCAS-AOD dataset contains 1510 images and 14 596 objects. There are two categories in this dataset, where the number of planes is 7482 objects and the number of the car is 7114. We randomly divide the dataset into 8:2 for training and testing. For all experiments on this dataset, we use the oriented annotations and scale the image to 512×512. We apply the same learning rate strategy as the HRSC2016 dataset.

B. Implementation Details

The backbone network of our FoRDet is VGG-16 [55] that is pretrained on the ILSVRC CLS-LOC dataset [56]. Different from RefineDet [20], we choose feature maps with the stride of 8, 16, 32, 64, and 128 pixels for prediction to handle the large-scale variation of geospatial objects. Table I shows the anchor scale settings over different feature maps. For the matching strategy, we set the foreground IoU threshold as 0.5 in both the coarse stage and the refined stage.

For the training phase, we use the stochastic gradient descent (SGD) method with a weight decay of 5e-4 and a momentum of 0.9 to optimize our network, and the batch size is set to 8 on a single 1080Ti GPU. Similar to [11] and [20], hard negative mining is used in the training phase, and random cropping, random expansion, and random image distortion are adopted as data augmentation strategies. In addition, we also use random rotation augmentation to increase the diversity of rotated objects.

For the testing phase, the score threshold and the IoU threshold in the rotation nonmaximum suppression (R-NMS) are set to 0.05 and 0.1, respectively. Besides, considering a large number of objects in RSIs, we preserve 2000 top-scored rotated bounding boxes per image.

C. Evaluation Metrics

In this article, we employ the average precision (AP) and the precision–recall curve (PRC) to evaluate the performance of the detector.

1) Precision–Recall Curve: PRC describes the relationship between precision and recall. The precision and the recall calculate the fraction of true positives and the fraction of positives that are correctly predicted. They are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{12}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

where TP, FP, and FN denote the numbers of true positives, false positives, and false negatives, respectively. Following [52], we set the IoU threshold as 0.5. The prediction is regarded as TP if the IoU between the prediction and the ground truth exceeds the threshold. Otherwise, the detection is regarded as an FP.

2) Average Precision: AP indicates the average value of precision in the interval of recall from 0 to 1 under a certain IoU threshold, where the precision (P) and the recall (R) calculate the fraction of true positives and the fraction of positives that are correctly predicted. The mAP denotes the mean of APs in all categories. The expressions are given in the following:

$$\text{AP} = \sum_{R \in (0, 0.1, \ldots, 1)} \frac{1}{11} P(R) \tag{14}$$

$$\text{mAP} = \frac{1}{C} \sum_{c=1}^{C} \text{AP}. \tag{15}$$

D. Ablation Study

To show the effectiveness of the proposed method, we conduct several ablation experiments on the DOTA validation set. For efficiency, we only train the model for 90k iterations in the ablation study, where the learning rate starts from 1e-3 and decreases to 1e-4 and 1e-5 at the 60k iterations and 75k iterations.

1) Baseline Setup: Inspired by the high-precision detection performance of the refined rotation structure in rotated object detection [19], we extend RefineDet [20] to a refine rotation structure as our baseline, termed R^2SSD. As shown in Table II, compared with our reimplementation Rotiaion-SSD, R^2SSD has a significant improvement in overall performance, especially on densely rotated arranged objects, such as SV, LV, and SP. Although the R^2SSD achieves good performance, it ignores the interference of the complex background in RSIs.

2) Effects of Foreground Relation Module: To strength the discriminative power of the foreground regions, we extend the baseline with the FRL module. As shown in the third row of Table II, the mAP increases from 66.06% to 67.85%. Furthermore, we also visualize the input feature map and the output feature map of the FRL module in Fig. 4. As depicted in Fig. 4, the background noise (e.g., grass and water) on the input feature map also has high activation response, which may interfere with the foreground regions. In contrast, the output feature map of the FRL module is concentrated on the foreground regions, which strengthens the discrimination of the foreground regions while suppressing the interference of background noise.

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TABLE II

ABLATION STUDY OF EACH PROPOSED COMPONENT IN OUR FoRDET ON THE DOTA VALIDATION SET. FRL AND FRW ARE THE SHORT NAMES OF FRL AND FRW LOSSES

Model	Iterations	mAP	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
R-SSD	90K	60.07	86.19	74.54	28.04	55.95	58.84	53.48	70.65	85.73	58.20	68.02	64.34	58.04	48.92	55.96	34.19
R^2SSD	90K	66.06	89.10	77.46	34.74	58.00	65.10	74.08	78.36	86.39	60.68	77.12	64.39	63.24	59.41	59.09	43.81
R^2SSD + FRL	90K	67.85	88.87	78.47	36.01	61.24	65.70	73.90	78.17	87.30	64.30	78.49	67.44	66.75	62.17	59.28	49.68
R^2SSD + FRW	90K	67.94	89.33	77.99	34.96	63.14	65.82	74.36	78.61	87.48	60.59	77.18	70.87	68.57	61.07	59.74	49.41
FoRDet	90K	69.09	89.41	78.69	37.18	63.46	66.55	74.44	78.57	88.04	65.68	79.34	72.80	**69.59**	62.31	59.27	51.06
FoRDet	180K	**72.35**	**89.50**	**85.50**	**40.53**	**72.37**	**69.44**	**76.92**	**79.55**	**89.45**	**72.74**	**79.98**	**76.44**	67.15	**64.25**	**60.76**	**60.75**

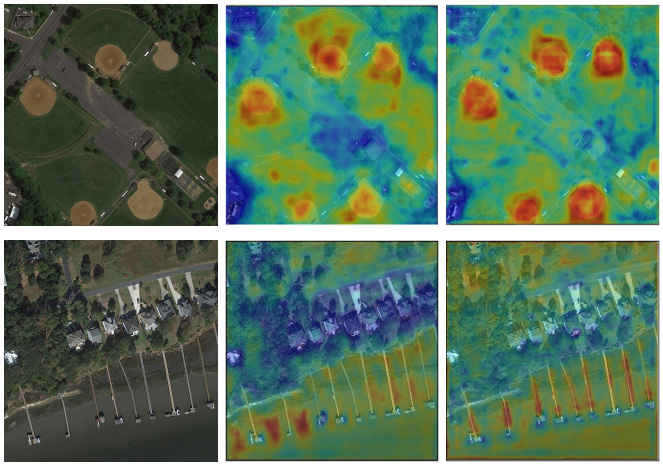


Fig. 4. Visualization of the input feature map (second column) and the output feature map (third column) at the conv4_3 level in the refined stage from R^2SSD + FRL.

3) Effects of Foreground Anchor Reweighting Loss: By employing the FRW loss to the baseline, the mAP has an improvement of 1.88%, as illustrated in the fourth row of Table II. It is worth noticing that the FRW loss significantly boosts the performance of GTF and SBF by more than 5%. We speculate that the GTF and SBF usually have a larger area and match a number of foreground anchors. As for the baseline, all foreground anchors are treated equally in the refined stage regardless of their classification confidence and location accuracy. However, the FRW loss adjusts the weights of different foreground anchors by calculating their foreground scores, which can make the model focus on potential foreground anchors. Fig. 5 demonstrates an example of different foreground anchors with the FRW loss. It can be seen that the anchor with the highest loss weight is more likely to contain foreground objects.

Furthermore, we also analyze the sensitivity of different hyperparameters used in the FRW loss. For the foreground score, α balances the importance between classification confidence (loc_a) and location accuracy (cls_a). As shown in Table III, more focus on loc_a can bring better performance improvement for the model. This is because loc_a is directly related to the ground truth, which can more accurately measure whether the foreground anchor contains the foreground region. Besides, we can find that it is better to integrate the classification confidence and location accuracy for the definition of the foreground scores than only consider from a single aspect.

TABLE III

ABLATION EXPERIMENTS ON THE SENSITIVITY OF HYPERPARAMETERS IN OUR FRW LOSS ON THE DOTA VALIDATION SET

Model	α	β	mAP
R^2SSD	-	-	66.01
R^2SSD + FRW	0	1	67.10
	0.25		67.24
	0.5		67.38
	0.75		**67.42**
	1		67.27
R^2SSD + FRW	0.75	1	67.42
		1.5	67.63
		2	**67.94**
		2.5	46.52
		3	-

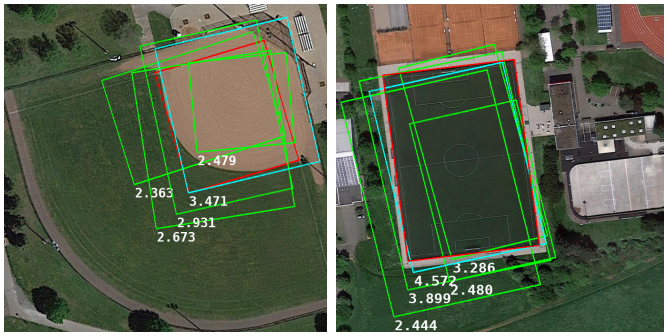


Fig. 5. Visualization of different foreground anchors with the FRW loss. For simplicity, we only show a small portion of foreground anchors. The red one represents the ground truth. The number in the lower left corner of each anchor is the adjusted weight in our FRW loss.

As for the reweighting factor, β controls the variance among different foreground anchors after reweighting. As shown in the seventh to ninth rows of Table III, $\beta = 2$ yields the best performance of 67.94%, which confirms a larger β offers better performance. However, when β is too large, the weights of the foreground anchors become unstable in the refined stage, leading to the model deteriorate seriously and even fails to converge.

4) Evaluation of FoRDet: After applying both the FRL module and the FRW loss to the baseline, we get our proposed FoRDet, and the overall performance increases to 69.09%, which indicates the complementarity between the two proposed modules. When we set the longer training iterations

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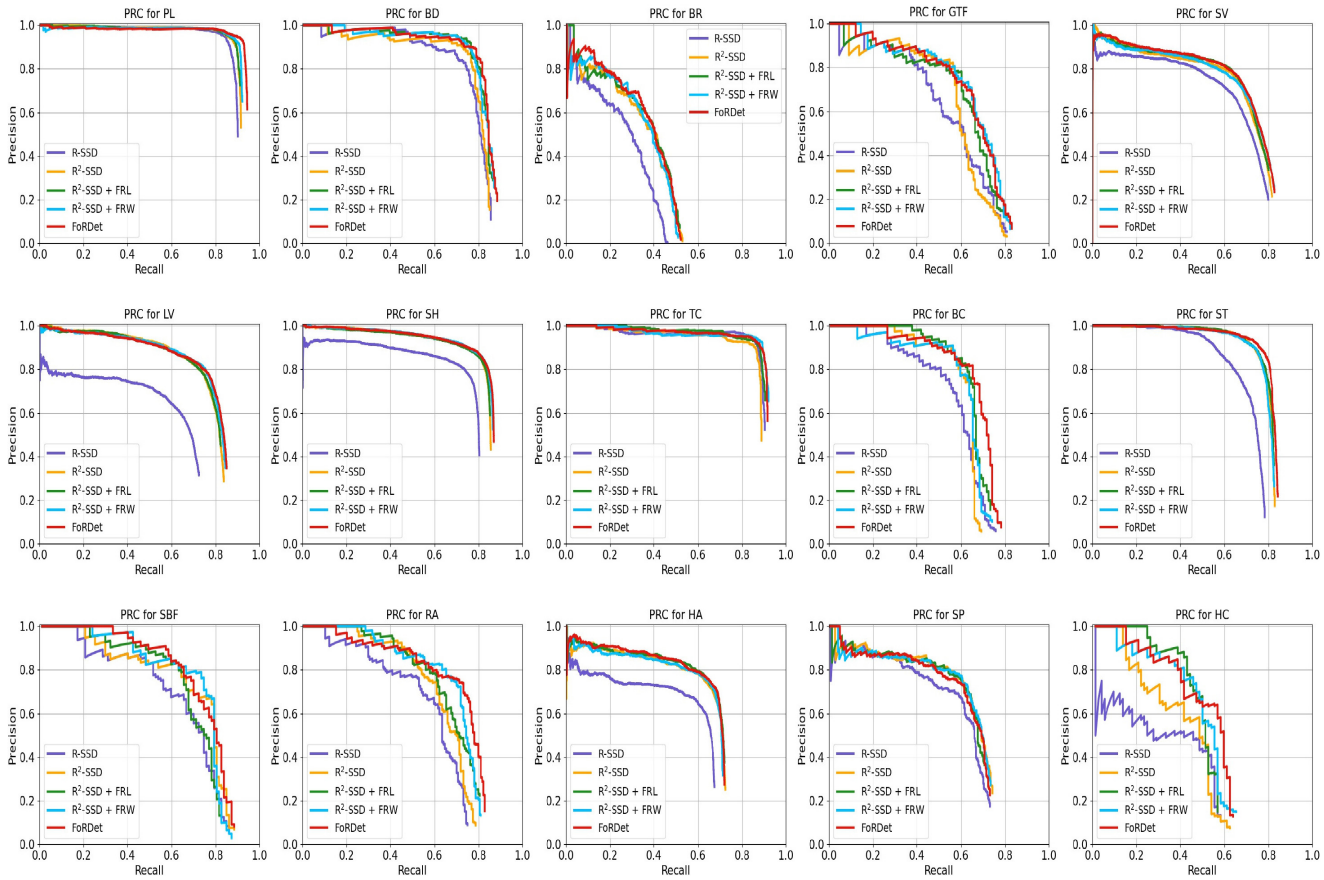


Fig. 6. PRCs of the proposed FoRDet (trained by 90k iterations) and other models on 15 categories of the DOTA validation set.

TABLE IV
COMPARISON RESULTS OF THE MAR ON THE DOTA VALIDATION SET

Model	R-SSD	R^2SSD	FoRDet
mAR	76.02	78.63	**80.54**

(i.e., 180k), the mAP can be further improved. Besides, Fig. 6 shows the PRCs on each category of the DOTA dataset.

In addition, we also use the mean Average Recall (mAR) to evaluate the effectiveness of the FoRDet. As shown in Table IV, the FoRDet obtains 4.52% and 1.91% performance gains, in terms of mAR compared with the R-SSD and R^2SSD, respectively. The above improvement lies in two aspects: 1) the refined stage in the refined rotation structure provides more accurate location regression and 2) the FRL module and FRW loss effectively improve the discrimination of the foreground regions and make the model focus on the potential foreground anchors.

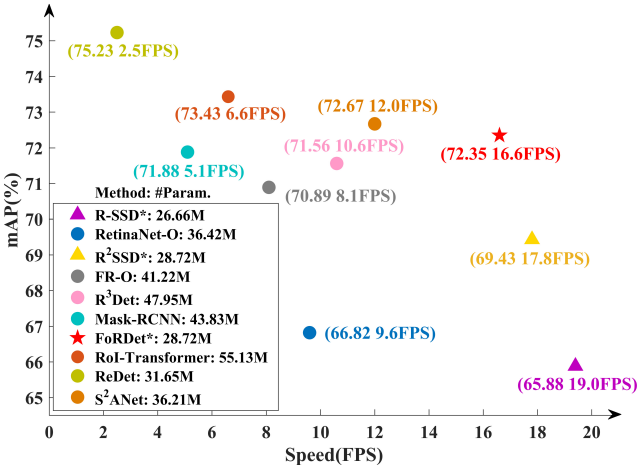


Fig. 7. Performance (mAP) versus speed (FPS) comparison on the DOTA validation set. We evaluate RetinaNet-O, FR-O, Mask-RCNN, R^3Det, RoI-Transformer, and S^2A-Net with ResNet-50. The ReDet uses ReResNet-50 as its backbone. Note that our reimplementations (R-SSD* and R^2SSD*) and our FoRDet* are all trained by 180k iterations with VGG16 as the backbones in this comparison.

E. Complexity and Speed Comparison

Apart from the above analysis, we also consider the complexity and speed comparison between the FoRDet and other approaches, as shown in Fig. 7. For the fair comparison, we evaluate the comparison approaches based on their open-source code bases[2][3][4] and follow their default settings. All detectors are trained on the DOTA training set, and the speed is measured on a single 1080Ti GPU. As shown

[2]https://github.com/dingjiansw101/AerialDetection
[3]https://github.com/SJTU-Thinklab-Det/r3det-on-mmdetection
[4]https://github.com/csuhan/s2anet https://github.com/csuhan/ReDet

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TABLE V

COMPARISONS PERFORMANCE ON THE DOTA TEST SET. † REPRESENTS TESTING WITH MULTISCALE. ‡ REPRESENTS MULTISCALE TRAINING AND MULTISCALE TESTING

Model	mAP	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
Two-stage																
R-DFPN [48]	57.94	80.92	65.82	33.77	58.94	55.77	50.94	54.78	90.33	66.34	68.66	48.73	51.76	55.10	51.32	35.88
R^2CNN [57]	60.67	80.94	65.67	35.34	67.44	59.92	50.91	55.81	90.67	66.92	72.39	55.06	52.23	55.14	53.35	48.22
RRPN [47]	61.01	88.52	71.20	31.66	59.30	51.85	56.19	57.25	90.81	72.84	67.38	56.69	52.84	53.08	51.94	53.58
ICN [14]	68.20	81.40	74.30	47.70	70.30	64.90	67.80	70.00	90.80	79.10	78.20	53.60	62.90	67.00	64.20	50.20
RoI-Trans.‡ [16]	69.56	88.64	78.52	43.44	75.92	68.81	73.6	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67
CAD-Net [58]	69.90	87.80	82.40	49.40	73.50	71.10	63.50	76.70	90.90	79.20	73.30	48.40	60.90	62.00	67.00	62.20
SCR-Det‡ [34]	72.61	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21
Gliding Vertex [17]	75.02	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	**70.91**	72.94	70.86	57.32
Mask-OBB‡ [59]	75.33	89.56	**85.95**	54.21	72.90	76.52	74.16	85.63	89.85	83.81	86.48	54.89	69.64	73.94	69.06	63.32
FFA‡ [49]	75.7	90.1	82.7	54.2	75.2	71.0	79.9	83.5	90.7	83.9	84.6	61.2	68.0	70.7	76.0	63.7
CenterMap-Net‡ [45]	76.03	89.83	84.41	54.60	70.25	77.66	78.32	87.19	90.66	84.89	85.27	56.46	69.23	74.13	71.56	66.06
ReDet‡ [60]	**80.10**	88.81	82.48	**60.83**	80.82	78.34	**86.06**	88.31	90.87	**88.77**	87.03	68.65	66.90	**79.26**	**79.71**	**74.67**
One-stage																
P-RSDet [29]	69.82	89.02	73.65	47.33	72.03	70.58	73.71	72.76	90.82	80.12	81.32	59.45	57.87	60.79	65.21	52.59
O^2-DNet [28]	71.04	89.31	82.14	47.33	61.21	71.32	74.03	78.62	90.76	82.23	81.36	60.93	60.17	58.21	66.98	61.03
DAL [44]	71.78	88.61	79.69	46.27	70.37	65.89	76.10	78.53	90.84	79.98	78.41	58.71	62.02	69.23	71.32	60.65
DRN† [31]	72.95	89.45	83.16	48.98	62.24	70.63	74.25	83.99	90.73	84.60	85.35	55.76	60.79	71.56	68.82	63.92
R^3Det‡ [19]	73.74	89.49	81.17	50.53	66.10	70.92	78.66	78.21	90.81	85.26	84.23	61.81	63.77	68.16	69.83	67.17
CSL‡ [61]	76.17	**90.25**	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93
S^2A-Net‡ [62]	79.42	88.89	83.60	57.74	**81.95**	79.94	83.19	**89.11**	90.78	84.87	**87.81**	70.30	68.25	78.30	77.01	69.58
FoRDet	71.98	89.67	83.21	36.96	69.61	75.46	73.96	78.61	**90.89**	85.41	77.56	63.71	62.36	67.17	72.57	52.62
FoRDet†	74.52	89.46	83.49	43.08	76.64	75.51	78.18	87.31	90.85	86.06	83.28	66.05	62.93	66.13	70.50	58.29
FoRDet‡	78.13	89.62	85.88	47.55	81.45	**80.63**	81.84	88.08	90.87	88.27	86.41	**72.42**	67.69	73.91	72.67	64.63

TABLE VI

COMPARISONS PERFORMANCE ON THE HRSC2016 TEST SET

Model	RRPN [47]	RRD [63]	RoI-Trans. [16]	Gliding Vertex [17]	R^3Det [19]	GRS-Det [30]	S^2A-Net [62]	ReDet [60]	FoRDet	FoRDet‡
mAP	79.05	84.30	86.20	88.20	89.26	89.57	90.17	**90.46**	89.92	90.23

in Fig. 7, the FoRDet has obvious advantages in complexity and speed while maintaining comparable performance compared to two-stage detectors (Mask-RCNN [64], RoI-transformer [16], and FR-O [52]). Compared with the state-of-the-art rotated object detection method (ReDet [60]), although our FoRDet has trailed its accuracy, the speed is around seven times faster than the speed of ReDet. In terms of the model complexity, our FoRDet requires fewer parameters than ReDet. Besides, our method is ahead of other one-stage methods (R-SSD, RetinaNet-O [13], R^2SSD, R^3Det [19], and S^2A-Net [62]) in performance, complexity, and speed. It is worth noting that the FoRDet preserves a similar amount of parameters (28 721 821 versus 28 721 776) with the baseline (R^2SSD) because the FoRDet only adds a small number of convolution and max-pooling layers in the refined stage.

F. Comparison With the State of the Art

1) Results on DOTA Dataset: Table V presents the performance of our method and the state-of-the-art methods on the OBB task of the DOTA test set. As shown in Table V, our FoRDet shows competitive performance at 71.98%, without bells and whistles, and outperforms many two-stage (ICN [14], RoI-Transformer [16], and CAD-Net [58]) and one-stage detectors (P-RSDet [29], O^2-DNet [28], and DAL [44]), which demonstrates the effectiveness of the proposed method. However, our method still has a big gap compared with other methods in the bridge. We speculate this is mainly because of the large aspect ratio of the bridge.

Due to the huge scale variations of objects in RSIs, the multiscale strategy can significantly improve the detection performance [16], [49], [59]. Thus, we resize the original image at three scales (1.0, 0.5, and 1.5) and generate the training and testing image patches with a size of 1024×1024 for the multiscale strategy. By employing the multiscale strategy in both the training and testing phases, our FoRDet achieves an overall performance of 78.13%. It is worth noticing that the BR, GTF, SH, and SBF obtain more than 10% performance gains with the multiscale strategy because of the immense interclass scale variation of these categories. Fig. 8 shows some qualitative results on the DOTA test set.

2) Results on HRSC2016 Dataset: Table VI depicts the comparison performance between our FoRDet with several state-of-art methods on the HRSC2016 test set. In comparison with the two-stage detectors, RoI-Transformer [16], and Gliding Vertex [17], our method obtains 3.72% and 1.72% performance gains, respectively, which reflects the superiority of the rotation refined structure. Although our FoRDet has comparable performance in terms of mAP, compared to

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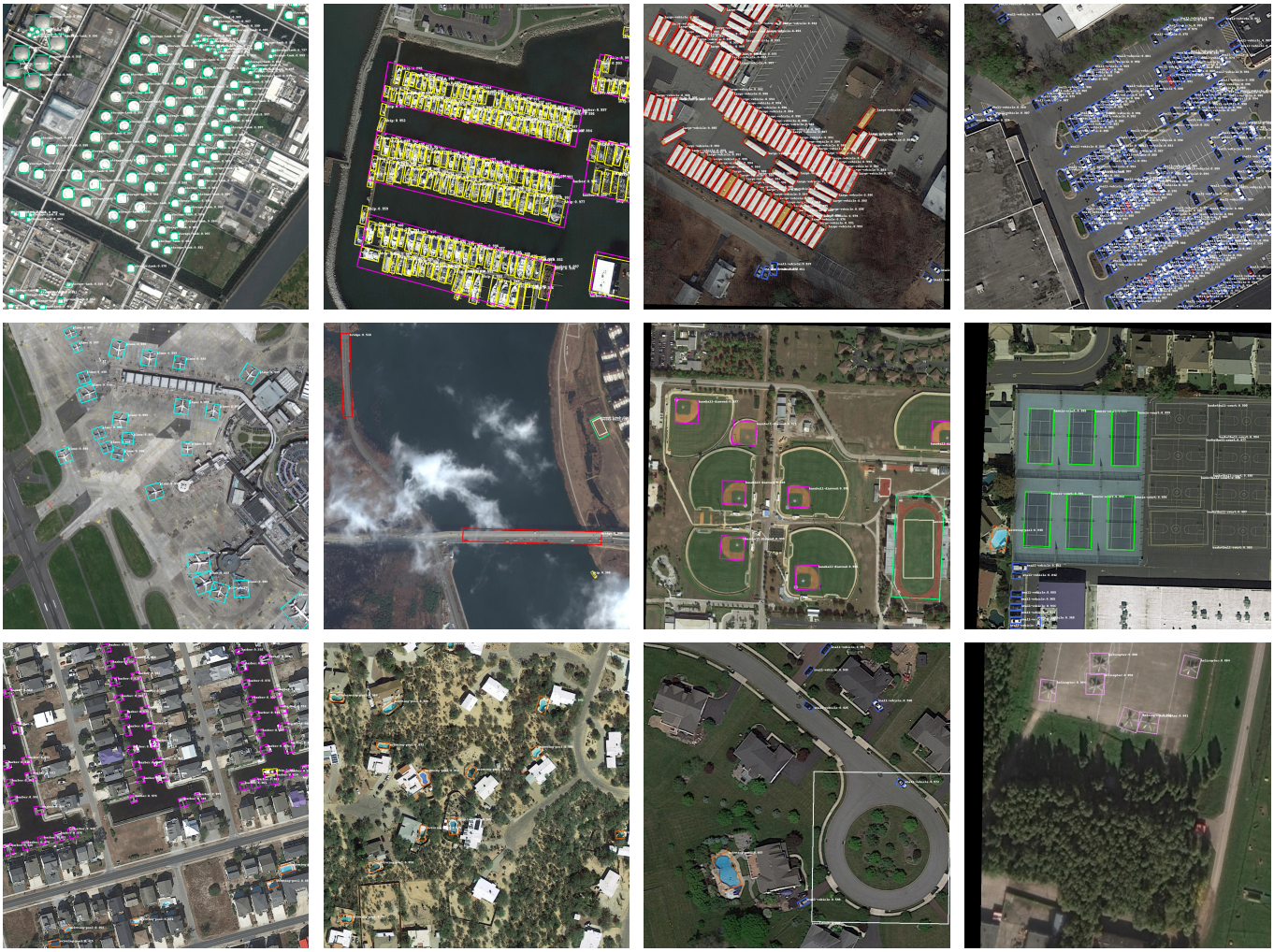


Fig. 8. Qualitative results from our proposed FoRDet on the DOTA test set.

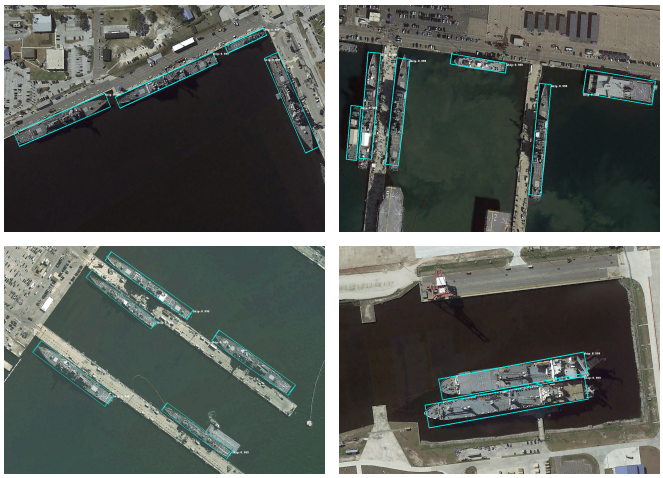


Fig. 9. Qualitative results from our proposed FoRDet on the HRSC2016 test set.



Fig. 10. Qualitative results from our proposed FoRDet on the UCAS-AOD test set.

GRS-Det [30], S^2A-Net [62], and ReDet [60], the FoRDet has a faster inference speed (FoRDet: 17.2 FPS, GRS-Det: 13.7 FPS, S^2A-Net: 13.0 FPS, and ReDet: 3.0 FPS). With the multiscale strategy, the proposed method achieves the mAP of 90.23%. Fig. 9 visualizes the detection results on the HRSC2016 test set.

3) Results on UCAS-AOD Dataset: We also evaluated the FoRDet on the UCAS-AOD dataset. As shown in Table VII, the FoRDet achieves an overall performance of 90.46%, which

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TABLE VII

COMPARISONS PERFORMANCE ON THE UCAS-AOD TEST SET. * REPRESENTS OUR REIMPLEMENTATIONS

Model	Plane	Car	mAP
RRPN [47]	88.04	74.36	81.20
R^2CNN [57]	89.76	78.89	84.32
R-DFPN [48]	88.91	81.27	85.09
DAL [44]	90.49	89.25	89.87
O^2-DNet [28]	**93.21**	86.72	89.96
P-RSDet [29]	92.69	87.38	90.03
S^2A-Net* [62]	90.74	89.88	90.31
ReDet* [60]	90.75	89.75	90.25
FoRDet	90.83	89.71	90.27
FoRDet‡	90.75	**90.17**	**90.46**

is better than other approaches. This superior performance confirms the effectiveness of our method once more. The detection results are sketched in Fig. 10.

V. CONCLUSION

In this article, we focus on the challenges of high background complexity in geospatial object detection. To alleviate the above problem, we introduce a rotation refined detector, called FoRDet. The proposed FoRDet not only enhances the discrimination of the foreground regions to reduce the background noise interference but also dynamically adjusts the weights of the foreground anchors in the loss functions to highlight the potential foreground anchors. Extensive experiments on different datasets show the effectiveness of our method.

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